Energy-weighted channel recovery in a fixed-topology JT model

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Abstract

We prove an energy-constrained classical-gravity recovery estimate for a specified one-particle scalar sector of a fixed-topology Jackiw–Teitelboim model. For the same initial gravitational Gaussian and every scalar input with mean mode excitation at most $\nu > 0$, including coherent infinite tails and arbitrary reference systems, the reduced channel approaches its scalar clock unitary at a sufficient rate $C^{-1/4}$. An explicit upper coefficient is $4.531256535935\sqrt{\nu}$. The proof keeps the energy budget inside a Gaussian Duhamel residual, including the direct charge-square term, before truncating the tail. A bounded second excitation moment gives a separate $C^{-1/2}$ estimate. The Hamiltonian, Gaussian moments, weighted charge inequalities, and domain argument are specified here; portable code checks the algebra and evaluates the bounds. No optimality, ADM-energy uniformity, or global novelty claim is made.

1 Specified model and result

Set the AdS radius and $\hbar$ to one. We use the fixed-topology canonical JT reduction with a free massless scalar and reflecting Dirichlet boundary conditions. The prior canonical framework is the physical Hilbert space and boundary generators of Penington and Witten [1], in particular their Eqs. (41)–(45). The scalar representation follows the positive-frequency Dirichlet branch [2]. On its one-particle sector, with orthonormal modes $|n\rangle$, $n \geq 0$, the charges are

$$K_0|n\rangle = (n+1)|n\rangle, \qquad K_+|n\rangle = \sqrt{(n+1)(n+2)}|n+1\rangle,$$
$$K_- = K_+^\dagger, \qquad K_2 = (K_+ - K_-)/(2i). \tag{1}$$

The modes are $\sin[(n+1)x]e^{-i(n+1)T}/\sqrt{\pi(n+1)}$ on global AdS_2, $0 < x < \pi$, with metric $(-dT^2 + dx^2)/\sin^2 x$. The charges preserve particle number. Restricting to this sector does not truncate its infinite mode ladder.

Write $C = \phi_b > 0$, $\chi = \log(2C) + q$, and $p = -i\partial_q$. Our physical simultaneous-boundary time generator is

$$H_C = \frac{p^2}{8C} + 2Ce^{2q} + e^q K_0 + \frac{K_2^2}{8C} \quad \text{on } L^2(\mathbb{R}) \otimes \ell^2(\mathbb{N}_0). \tag{2}$$

The cited boundary sum is $2C(H_L + H_R) = p_\chi^2/4 + K_2^2/4 + e^\chi K_0 + e^{2\chi}$, with $J_1 = -K_0$, $J_3 = -K_2$; hence (2) is $H_L + H_R$ after this shift. It is not a modified decoupled Hamiltonian. We use the positive self-adjoint realization specified in Section 5.

The normalized initial gravity wavefunction and reference scalar clock are

$$G_C(q,0) = \left(\frac{4C}{\pi}\right)^{1/4} e^{-2Cq^2}, \qquad U_{\mathrm{cl}}(t) = e^{-i\theta(t)K_0}, \quad \theta(t) = \arctan(\sinh t). \tag{3}$$

Let $\Phi_C(t)$ adjoin this gravity state, evolve with H_C, and trace gravity, and let $\Phi_{\mathrm{cl}}(t) = \mathrm{Ad}_{U_{\mathrm{cl}}(t)}$. Define the unhalved energy-constrained distance

$$d_\nu(C,T) = \sup_{\substack{0\le t\le T,\ R,\ \rho_{mR}\ge 0\\ \mathrm{Tr}\,\rho_{mR}=1,\ \mathrm{Tr}(n\rho_m)\le\nu}} \left\|[(\Phi_C(t)-\Phi_{\mathrm{cl}}(t))\otimes \mathrm{Id}_R](\rho_{mR})\right\|_1, \quad n = K_0 - 1. \tag{4}$$

Here R ranges over reference systems. No diagonality, finite second moment, or finite ADM mean is assumed.

Put

$$\begin{aligned} M(v) &= e^{8v} - 2e^{2v}(1+6v+8v^2) + 1 + 8v + 12v^2, \\ F(v) &= e^{2v} - 2e^{v/2} + 1, \qquad g_C = 2C\sqrt{M(1/(8C))}. \end{aligned} \tag{5}$$

For integer $n \ge 0$ define the nonnegative coefficient polynomials

$$u(n) = \frac{(n+1)(n+2)^2(n+3)}{16}, \quad d(n) = \frac{(n+1)^4}{4}, \quad \ell(n) = \frac{n^2(n^2-1)}{16}, \tag{6}$$

and, for $N \ge 1$,

$$B_N(\nu) = \sqrt{\frac{3}{4} + \frac{u(N)-3/4}{N}\nu} + \sqrt{\frac{1}{4} + \frac{d(N)-1/4}{N}\nu} + \sqrt{\frac{\ell(N)}{N}\nu}. \tag{7}$$

Theorem 1. *For every integer $N \ge 1$, $C > 0$, $\nu \ge 0$, and $0 \le T \le 1$,*

$$\begin{aligned} d_\nu(C,T) \le \min\Bigg\{ 2,\ 2T\left[g_C + \sqrt{1+(N+2)\nu}\sqrt{F(1/(8C))} + \frac{B_N(\nu)}{8C}\right] \\ + 4\sqrt{\frac{\nu}{N+1}}\Bigg\}. \end{aligned} \tag{8}$$

For each fixed $\nu > 0$, choosing $N = \lfloor (4\sqrt{2}/3)\sqrt{C}\rfloor$ gives

$$d_\nu(C,1) \le \frac{14\,2^{3/4}\sqrt{3}}{9}\sqrt{\nu}\,C^{-1/4} + O_\nu(C^{-1/2}). \tag{9}$$

For $\nu = 0$, the input is the ground mode and $d_0(C,T) \le 2T[g_C + \sqrt{F(1/(8C))} + 1/(8C)] = O(C^{-1/2})$.

2 Normalized Gaussian reference and its residual

The reference gravitational path and stability solution are

$$q_c = -\log\cosh t, \quad p_c = -4C\tanh t, \quad z(t) = 1 - t\tanh t + i\tanh t. \tag{10}$$

They obey $\dot q_c = p_c/(4C)$, $\dot p_c = -4Ce^{2q_c}$, $z'' + 2\,\mathrm{sech}^2(t)z = 0$, $z(0) = 1$, and $z'(0) = i$. With $\xi = q - q_c(t)$ and $S_C(t) = 2C[t - 2\tanh t]$, the Gaussian

$$G_C(q,t) = \left(\frac{4C}{\pi}\right)^{1/4} z(t)^{-1/2}\exp\left(i\left[S_C + p_c\xi + 2C\frac{z'}{z}\xi^2\right]\right) \tag{11}$$

uses the square root continuous from $z(0) = 1$. The Wronskian gives $\mathrm{Im}(z'/z) = |z|^{-2}$; thus it is normalized and its position variance is $v_C(t) = |z(t)|^2/(8C)$. Substitution shows that it solves

$$H_g^{(2)}(t) = \frac{p^2}{8C} + 2Ce^{2q_c(t)}(1 + 2\xi + 2\xi^2). \tag{12}$$

The phase in (11) includes the constant term of this quadratic Taylor Hamiltonian. Writing $y = \tanh t$ gives $|z|^2 - 1 = y[(1+t^2)y - 2t] \le 0$ on $[0,1]$, since $y \le t$ and $1 + t^2 \le 2$.

For a finite-sequence scalar/reference vector f, set $f(t) = (U_{\mathrm{cl}}(t) \otimes I)f$. The product $\Psi_{\mathrm{app}} = G_C(t) \otimes f(t)$ has exact residual

$$\begin{aligned}(H_C - i\partial_t)\Psi_{\mathrm{app}} = 2Ce^{2q_c}(e^{2\xi} - 1 - 2\xi - 2\xi^2)G_C \otimes f(t) \\ + e^{q_c}(e^{\xi} - 1)G_C \otimes K_0 f(t) + G_C \otimes \frac{K_2^2 f(t)}{8C}.\end{aligned} \tag{13}$$

All three terms are retained. In particular the direct charge-square term is not replaced by a scalar expectation value.

For a centered normal variable X of variance v, $\mathbb{E}e^{aX} = e^{a^2 v/2}$. Differentiating this identity gives

$$\begin{aligned}\mathbb{E}[e^{2X}(1 + 2X + 2X^2)] = e^{2v}(1 + 6v + 8v^2), \\ \mathbb{E}(1 + 2X + 2X^2)^2 = 1 + 8v + 12v^2.\end{aligned} \tag{14}$$

Consequently the two squared residual norms are exactly

$$\mathbb{E}(e^{2X} - 1 - 2X - 2X^2)^2 = M(v), \qquad \mathbb{E}(e^X - 1)^2 = F(v). \tag{15}$$

Both functions increase for $v \ge 0$. All nonconstant coefficients of F are positive; those of M vanish below degree three and at degree $k \ge 3$ equal $2^k[4^k - 2(2k^2 + k + 1)]/k! > 0$. Their leading terms are $M(v) = (80/3)v^3 + O(v^4)$ and $F(v) = v + O(v^2)$. Since $e^{q_c}, e^{2q_c} \le 1$, Duhamel's identity bounds the vector error, uniformly for $t \le T \le 1$, by

$$T\left[g_C \|f\| + \sqrt{F(1/(8C))}\,\|K_0 f\| + \frac{\sup_{s \le T}\|K_2^2 f(s)\|}{8C}\right]. \tag{16}$$

This compares exact and reference evolutions from the same initial vector. The scalar phase preserves mode probabilities and $\|K_0 f\|$.

3 Energy-weighted charge bound and coherent tail

Purify an input of (4), enlarging its reference if needed. Let $P_N = \sum_{n=0}^{N} |n\rangle\langle n|$ and $f = (P_N \otimes I)\psi$. Then $\|f\| \le 1$ and, with reference-valued coefficients f_n,

$$\sum_{n=0}^{N} n\,\|f_n\|^2 \le \nu, \qquad \|K_0 f\|^2 \le 1 + (N+2)\nu. \tag{17}$$

The second inequality uses $(n+1)^2 = 1 + n(n+2)$ and $n \le N$. Squaring the *infinite* charge in (1) gives

$$\begin{aligned}K_2^2|n\rangle = &-\frac{(n+2)\sqrt{(n+1)(n+3)}}{4}|n+2\rangle + \frac{(n+1)^2}{2}|n\rangle \\ &- \frac{n\sqrt{(n-1)(n+1)}}{4}|n-2\rangle,\end{aligned} \tag{18}$$

where the downward term is zero at $n = 0, 1$. Its three squared coefficients are (6). Input through mode N can produce output $N + 2$; squaring a truncated charge would omit this contribution.

Lemma 1. *If $\|f\| \le 1$, $f \in \mathrm{Ran}(P_N \otimes I)$, and $\sum n\,\|f_n\|^2 \le \nu$, then $\|K_2^2 f\| \le B_N(\nu)$.*

Proof. For integer $0 \le n \le N$,

$$u(n) \le \tfrac{3}{4} + \frac{n}{N}[u(N) - \tfrac{3}{4}], \qquad d(n) \le \tfrac{1}{4} + \frac{n}{N}[d(N) - \tfrac{1}{4}], \qquad \ell(n) \le \frac{n}{N}\ell(N). \tag{19}$$

The first two quotients after subtracting constants are

$$\frac{u(n) - 3/4}{n} = \frac{n^3}{16} + \frac{n^2}{2} + \frac{23n}{16} + \frac{7}{4}, \qquad \frac{d(n) - 1/4}{n} = \frac{n^3}{4} + n^2 + \frac{3n}{2} + 1, \tag{20}$$

increasing for $n \ge 0$. The last is $(n^3 - n)/16$, increasing at integers $n \ge 1$, and $\ell(0) = 0$. Each of the three shifts in (18) has squared norm equal to its coefficient polynomial averaged against $\|f_n\|^2$. Apply (19), then the triangle inequality between shifts. This gives (7) for reference-valued coefficients. □

Proof of Theorem 1. Equation (16), (17), and Lemma 1 bound the projected vector error by the bracket in (8) times T. These bounds hold at all reference times because phases preserve mode probabilities. Both Stinespring vectors have norm $\|f\| \le 1$, so their reduced subnormalized density matrices differ in trace norm by at most twice their vector distance.

The omitted probability obeys $p = \|(1 - P_N)\psi\|^2 \le \nu/(N+1)$. In the span of the retained and omitted components, the pure-state density minus its subnormalized projection has trace norm

$$\||\psi\rangle\langle\psi| - |f\rangle\langle f|\|_1 = \sqrt{4p - 3p^2} \le 2\sqrt{p}. \tag{21}$$

Partial tracing preserves the upper bound. Both channels contract Hermitian trace norm; their difference contributes at most $4\sqrt{p}$ on the omitted part. This proves (8), uniformly over references, without differentiating an infinite-tail vector.

For fixed $\nu > 0$, $B_N(\nu) = \sqrt{\nu}N^{3/2} + O_\nu(N^{1/2})$. With $N = \lfloor a\sqrt{C} \rfloor$, the leading coefficient is

$$\sqrt{\nu}\left(\sqrt{a/2} + \frac{a^{3/2}}{4} + \frac{4}{\sqrt{a}}\right) C^{-1/4}. \tag{22}$$

The gravity term is only $2g_C = \sqrt{5/6}\, C^{-1/2} + O(C^{-3/2})$. The leading coefficient is stationary at $1/\sqrt{2} + 3a/4 - 4/a = 0$, with positive solution $a = 4\sqrt{2}/3$. This proves (9). For $\nu = 0$ the input class is exactly the ground mode and $\|K_0|0\rangle\| = \|K_2^2|0\rangle\| = 1$. □

4 Bounded second moment: a separate input class

If $\mathrm{Tr}(n^2\rho_m) \le \mu$, the projected vector satisfies $\mathbb{E}n \le \sqrt{\mu}$ and $\mathbb{E}n^j \le N^{j-2}\mu$ for $j = 2, 3, 4$. Thus $\|K_0 f\| \le 1 + \sqrt{\mu}$ and

$$\begin{aligned} B_N^{(2)}(\mu) = {} & \sqrt{\tfrac{3}{4} + \tfrac{7}{4}\sqrt{\mu} + (N^2/16 + N/2 + 23/16)\mu} \\ & + \sqrt{\tfrac{1}{4} + \sqrt{\mu} + (N^2/4 + N + 3/2)\mu} + \tfrac{1}{4}N\sqrt{\mu} \end{aligned} \tag{23}$$

bounds $\|K_2^2 f\|$. Expand u, d, ℓ and use the same shift triangle inequality to obtain this expression. The tail probability is now at most $\mu/(N+1)^2$.

Corollary 1. *For the distance with the stronger constraint* $\mathrm{Tr}(n^2\rho_m) \le \mu$*, make the following replacements in* (8)*:*

$$\sqrt{1 + (N+2)\nu} \mapsto 1 + \sqrt{\mu}, \quad B_N(\nu) \mapsto B_N^{(2)}(\mu), \quad 4\sqrt{\nu/(N+1)} \mapsto \frac{4\sqrt{\mu}}{N+1}.$$

Taking $N = \lfloor 4\sqrt{C} \rfloor$ *gives*

$$d_{\mu;2}(C, 1) \le \left[\sqrt{5/6} + \frac{1 + \sqrt{\mu}}{\sqrt{2}} + 2\sqrt{\mu}\right] C^{-1/2} + O_\mu(C^{-1}). \tag{24}$$

No fourth moment of the full input is required: only its finite projection enters Duhamel, and bounded exact channels act on the tail.

5 Domains, quantitative consequences, and limits

The realization of (2) is associated with the closed nonnegative form

$$\mathfrak{h}_C[\psi] = \frac{\|p\psi\|^2 + \|K_2\psi\|^2}{8C} + 2C\,\|e^q\psi\|^2 + \left\|e^{q/2}K_0^{1/2}\psi\right\|^2. \tag{25}$$

Its domain is the intersection of the four indicated operator domains. The Jacobi charge has its standard self-adjoint closure; its linearly growing coefficients satisfy the divergent reciprocal-coefficient criterion. The finite sum of closed nonnegative forms on this dense intersection is closed. Smooth compact geometry functions times finite scalar sequences form a common form core. To see this, first use smooth scalar cutoffs $f(n/N)$: the commutator $[K_2, f(n/N)]$ is uniformly bounded and supported in a moving tail, hence tends strongly to zero on every vector. These cutoffs converge in the K_2 graph norm and commute with K_0 and geometry operators. Smooth compact geometry cutoffs then have vanishing derivative commutators; smoothing on compact support completes the approximation. This does not assume sharp scalar projections converge in an unbounded graph norm.

For the finite scalar inputs used in Duhamel, applying the differential expression to the Gaussian reference produces finite sums of Gaussian polynomials times exponentials and finite scalar sequences. Every derivative and exponential multiple in (25) is square integrable. Integration by parts against the common form core therefore puts these trial vectors in the operator domain, with action (13). Their time dependence is continuously differentiable on the compact interval, so Duhamel's identity is valid. General energy-constrained inputs enter through the trace-norm tail step; they need not be in $\operatorname{Dom} H_C$.

Table 1 gives outward-rounded 150-decimal interval bounds. The earlier worst-code projection argument substitutes $N+1$ and $(N+1)(N+2)$ for the weighted norms; $N = \lfloor (8\nu C)^{1/3} \rfloor$ then gives $3\sqrt{2}\nu^{1/3}C^{-1/6} + O(C^{-1/3})$. The new bound retains the mean budget inside the residual, rather than re-optimizing the same worst-code estimate. At $N \asymp \sqrt{C}$ its constrained vector error vanishes although the worst-code operator estimate need not. For fixed ν, the sufficient small-error resource scaling becomes $C = O(\nu^2\delta^{-4})$, compared with $O(\nu^2\delta^{-6})$ from that earlier argument. These are sufficient asymptotics, not necessary thresholds.

Table 1: Unhalved channel-distance upper bounds at $\nu = 1$, $T = 1$; every displayed value is rounded upward.

C	Energy-weighted bound	Worst-code projection bound
10^6	0.144226	0.435331
10^{12}	0.004532171	0.042528
10^{18}	0.000143292	0.004244
10^{24}	0.000004531258	0.000424275

The portable script checks exact charge and Gaussian-moment identities, the stability equation and Wronskian, and charge columns retaining two upper-edge raising steps. Its 180 coherent matter/reference samples support the inequalities; arbitrary-reference coverage follows from the proof. As a hostile tail check, $\sqrt{1-1/N}|0\rangle + \sqrt{1/N}|N\rangle$ has mean excitation one, but $\|K_0\psi\|^2 = N+3$ and projection error $\sqrt{4/N - 3/N^2}$. A mean-as-norm substitution or probability-only coherent-tail bound fails.

The constraint is scalar global-mode excitation, not total ADM energy; finite mean K_0 alone need not imply finite K_2 form energy. This is leading scalar-clock recovery, distinct from a quantum-corrected finite-code channel expansion. No optimal exponent, arbitrary-time, topology-changing, general-Fock, boundary-instrument, or four-dimensional GR conclusion follows. The computation evaluates explicit bounds and finite coefficient checks; it does not simulate the full JT propagator.

Provenance and disclosure

The canonical JT framework cited above is prior work. The Gaussian comparison was inherited from an internal September 2026 calculation; the energy-weighted improvement was derived on 28 October 2026 and is packaged here on 29 October 2026. Historical source dates, this delivery, and demonstrable public disclosure are distinct: a private commit does not establish public priority. No global novelty certification is claimed.

Earlier work was Fable 5-led; July 2026 onward results were predominantly produced by GPT models, as reported by the operator. DeepSeek and multiple GPT models also contributed cross-review and manuscript preparation. This is reported chronology, not a per-result authorship allocation. Human names and affiliations are blank, and human expert review has not occurred. The supplied statement is preserved verbatim:

> "I genuinely do not know how authorship, intellectual credit, or ownership should be attributed for these results."

References

[1] Geoff Penington and Edward Witten. Algebras and states in JT gravity, 2023. URL https://arxiv.org/abs/2301.07257v3. Version 3; canonical reduction in Eqs. (41)–(45).

[2] Atsushi Higuchi, Lasse Schmieding, and David Serrano Blanco. Scalar field in AdS_2 and representations of $\widetilde{SL}(2,\mathbb{R})$. *Journal of Mathematical Physics*, 63:122301, 2022. doi: 10.1063/5.0117631. URL https://arxiv.org/abs/2107.08754v3.